Biomedical Materials:

Bioceramics-1

Assoc. Prof. Dr. Ali Can Zaman

Aim

To know bioactive glasses

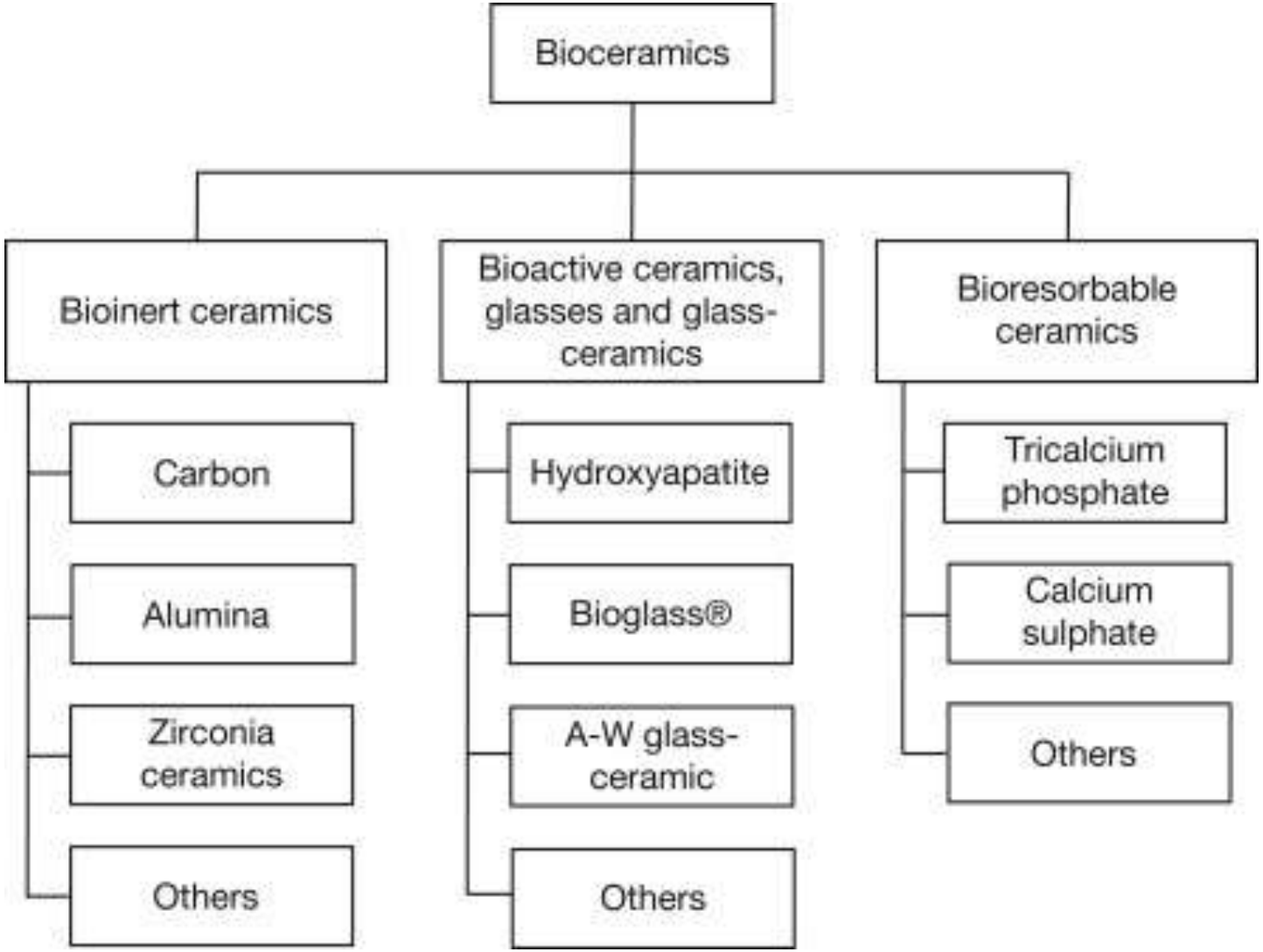
To know glass ceramics

To know hydroxyapatite

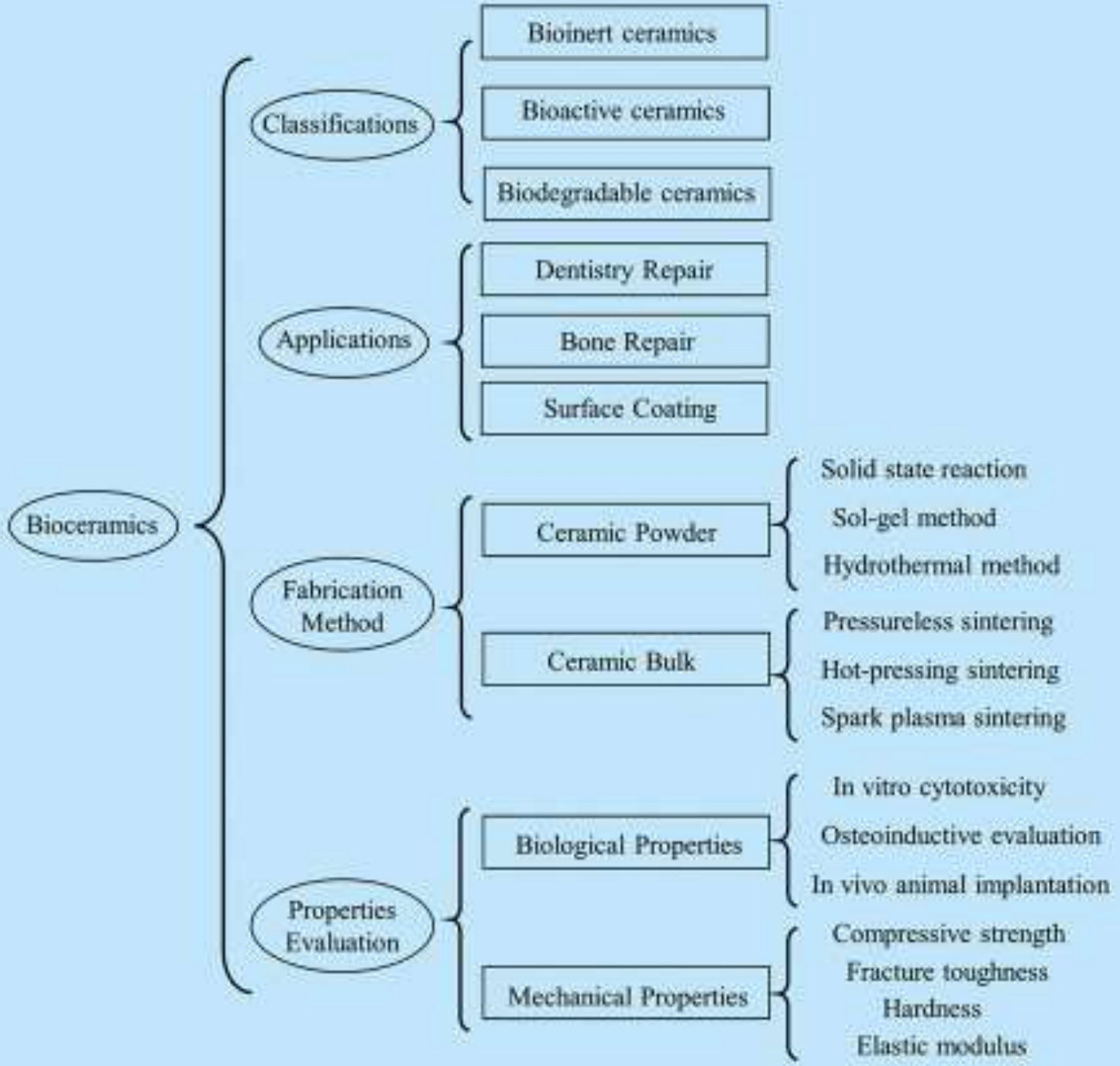
To know calcium phosphate, calcium sulfate

No material implanted in living tissue is completely inert because all materials elicit a response from living tissues. Nearly-inert bioceramics tend to be used where a robust hard surface is needed, e.g. a joint replacement with articulating bearing surface or dental restoration. Nearly-inert implants are commonly surrounded by fibrous capsules in the body due to the body's inflammatory response to foreign objects. Bioactivity is regarded as the provocation of a beneficial biological response from the body's host environment to the implanted devices. In the case of bioactive ceramics, glasses, and glass-ceramics, this has traditionally been regarded as the ability to form a direct bond with bone tissue without fibrous encapsulation.

Classification of bioceramics



Classification



Classification

a) Bioinert Ceramic: *Alumina (Aluminum Oxide) Dental Implant (Al_2O_3)*
Alumina is bioinert, meaning it does not react with biological tissues, making it ideal for dental implants and other applications where inertness is required.

Classification

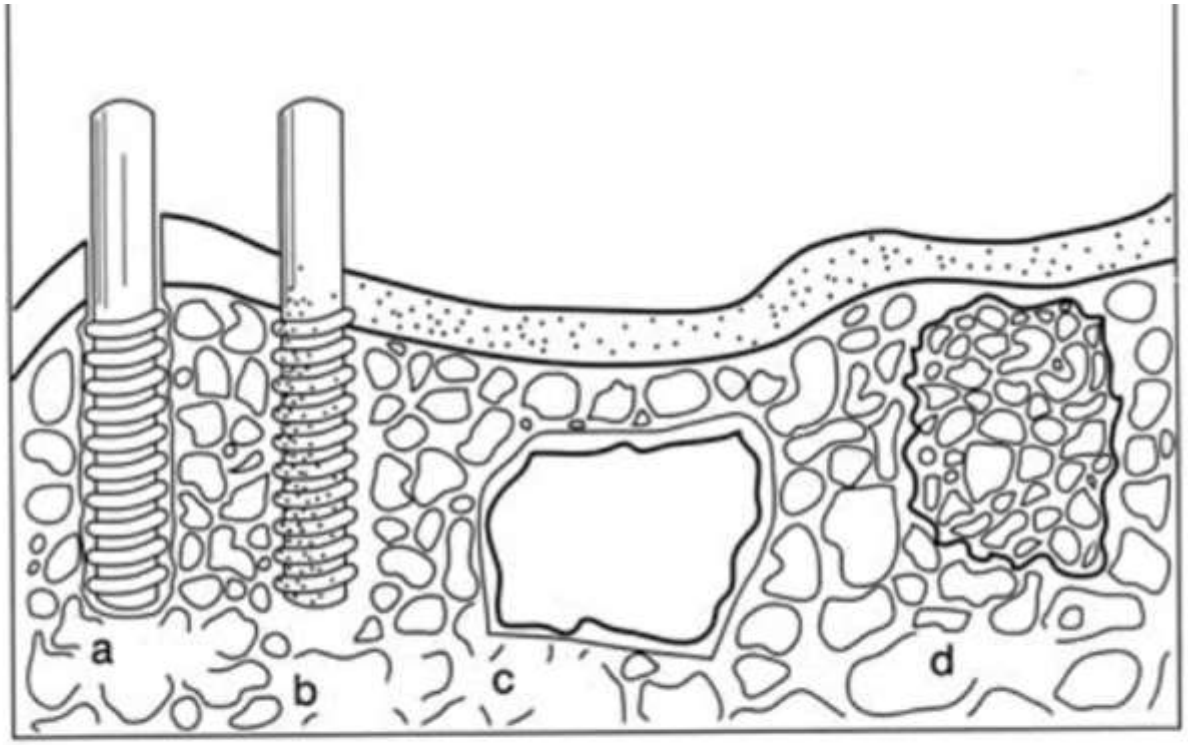
b) Bioactive Ceramic:

Hydroxyapatite Coating on Metal ($Ca_{10}(PO_4)_6(OH)_2$)
Hydroxyapatite is bioactive, which allows it to bond well with bone tissue. It is often used as a coating on metal implants to enhance biocompatibility and encourage bone integration.

Bioglass: This glass-ceramic material releases ions that stimulate cellular responses, encouraging new bone formation around the implant. Over time, some parts of Bioglass may dissolve, integrating with the surrounding tissue, but significant portions of it can remain intact as a structural component. (Bioglass releases ions (like calcium, sodium, and phosphorus) that help stimulate the formation of a layer similar to hydroxyapatite on its surface, which encourages bone bonding.)

Classification

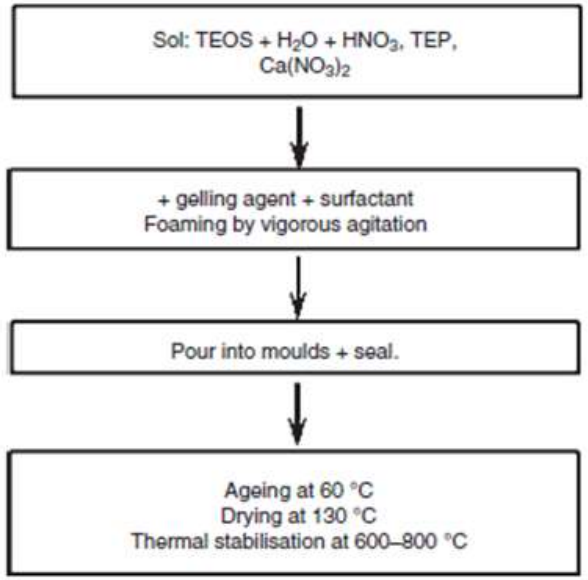
d) Biodegradable Ceramic: *Tricalcium Phosphate* ($Ca_3(PO_4)_2$)
Tricalcium phosphate is biodegradable, which allows it to be gradually absorbed by the body, making it suitable for applications where temporary support is required, such as bone repair scaffolds.



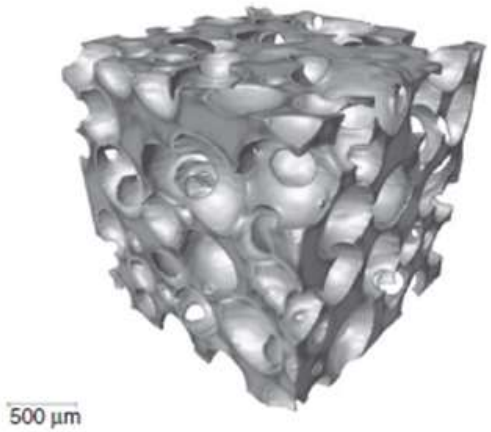
Classification of bioceramics according to their bioactivity; (a) bioinert, alumina dental implant, (b) bioactive, hydroxyapatite (Ca 10 (PO 4) 2 (OH) 2)coating on a metallic dental implant, (c) surface active, bioglass, (d) bioresorbable tri-calcium phosphate implant (Ca 3 (PO 4) 2).

Bioactive glasses





12.3 A flow diagram of the sol–gel foaming process.



12.4 X-ray microtomography (µCT) image of a typical bioactive glass scaffold produced by the sol–gel foaming process. Image: Gowsihan Poologasundarampillai.

Bioactive glasses

Bioactive Glasses (Bioglass) Forming Hydroxycarbonate Apatite (HCA): Bioactive glasses are known to form an HCA layer on their surface when in contact with physiological fluids, which allows them to bond with bone tissue. This HCA layer resembles the mineral component of bone and is key to the bioactivity and bone bonding of bioglasses.

Mechanism of Bioactive Bonding: bioactive glass achieves bonding through the exchange of surface silica groups with calcium (Ca) and phosphorus (P). This ion exchange reaction contributes to the formation of the HCA layer, facilitating the integration between the implant and surrounding bone tissue.

Disadvantages (Amorphous Structure and Mechanical Weakness): Bioactive glasses are often amorphous, which can lead to lower mechanical strength compared to crystalline materials. This mechanical weakness is a disadvantage, limiting their use in load-bearing applications.

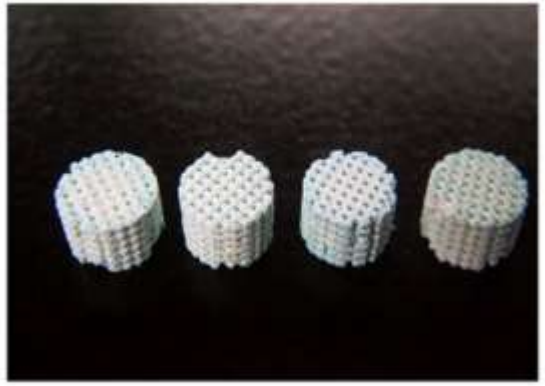
Bioactive glasses

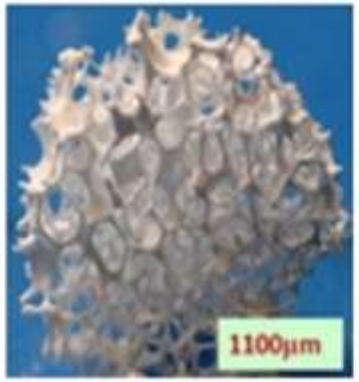
Composition Differences from Commercial Glass:

Lower SiO_2 Content (< 60%): Bioactive glasses typically have a lower silica content compared to traditional commercial glasses, which allows for better ion release and bioactivity.

Higher Na_2O and CaO Content: Increased sodium oxide (Na_2O) and calcium oxide (CaO) in the composition enhance the reactivity of bioactive glasses with body fluids.

Higher CaO/P_2O_5 Ratio: This ratio promotes the formation of an HCA layer, which is essential for bonding with bone.







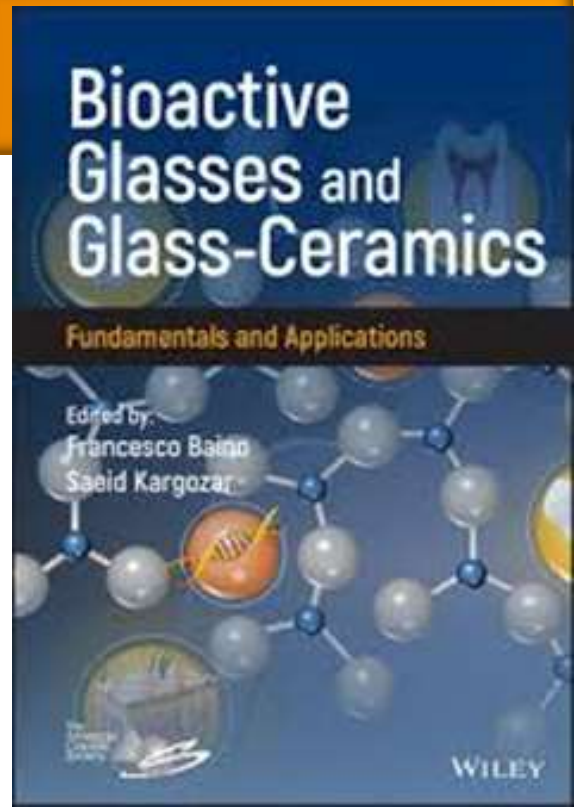
Ticari camlardan bileşim açısından farkları,
- ✓ düşük SiO_2 (<% 60),
- ✓ yüksek Na_2O ile CaO ve
- ✓ yüksek CaO/P_2O_5 oranıdır (5)

45S5 is the first bioglass invented (1969).
It is calcium sodium phosphosilicate.

Composition of 45S5

SiO_2	CaO	Na_2O	P_2O_5
45	24,5	24,5	6

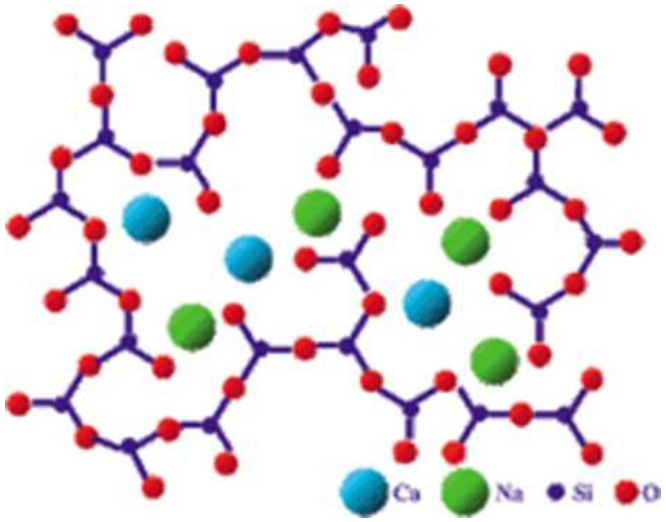
The structure of glass



In bioactive glasses, SiO_2 (silicon dioxide) serves as the main glass-forming component, creating the glass network structure.

Compounds like SrO (strontium oxide), MgO (magnesium oxide), CaO (calcium oxide), and Na_2O (sodium oxide) act as network modifiers. These components lower the melting temperature of silica (from around 1700 °C to 1500 °C or lower), making it easier to process the glass.

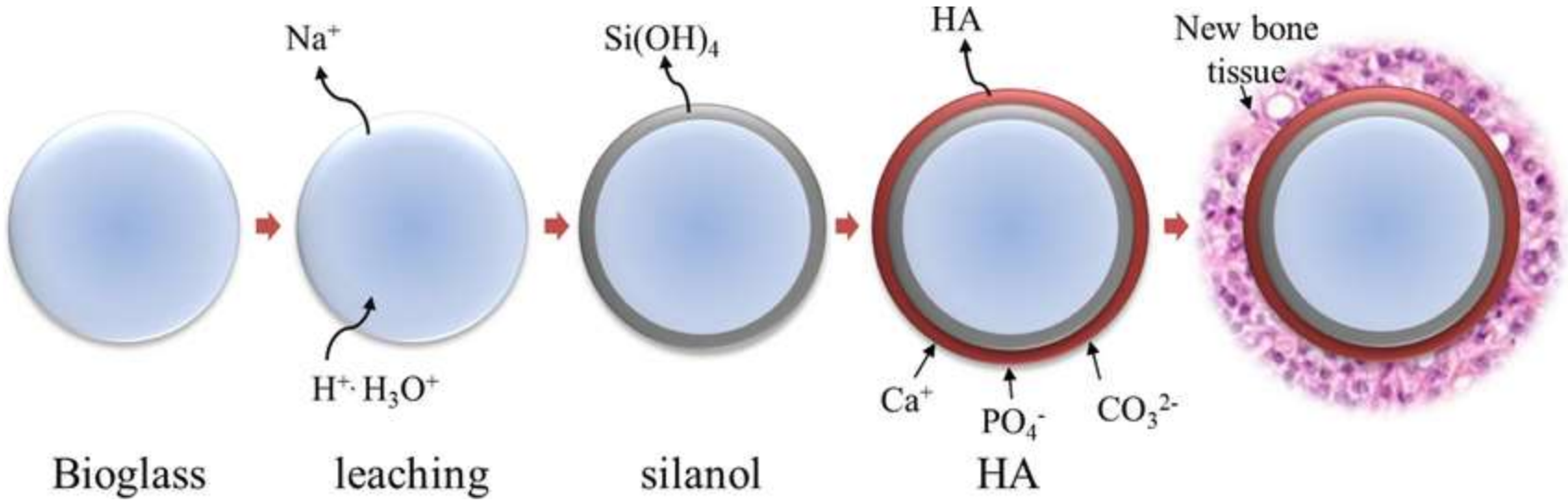
Intermediate oxides can act as network modifiers or as part of the network, depending on the composition and production technique.



How does bioactive glass interacts with tissues?

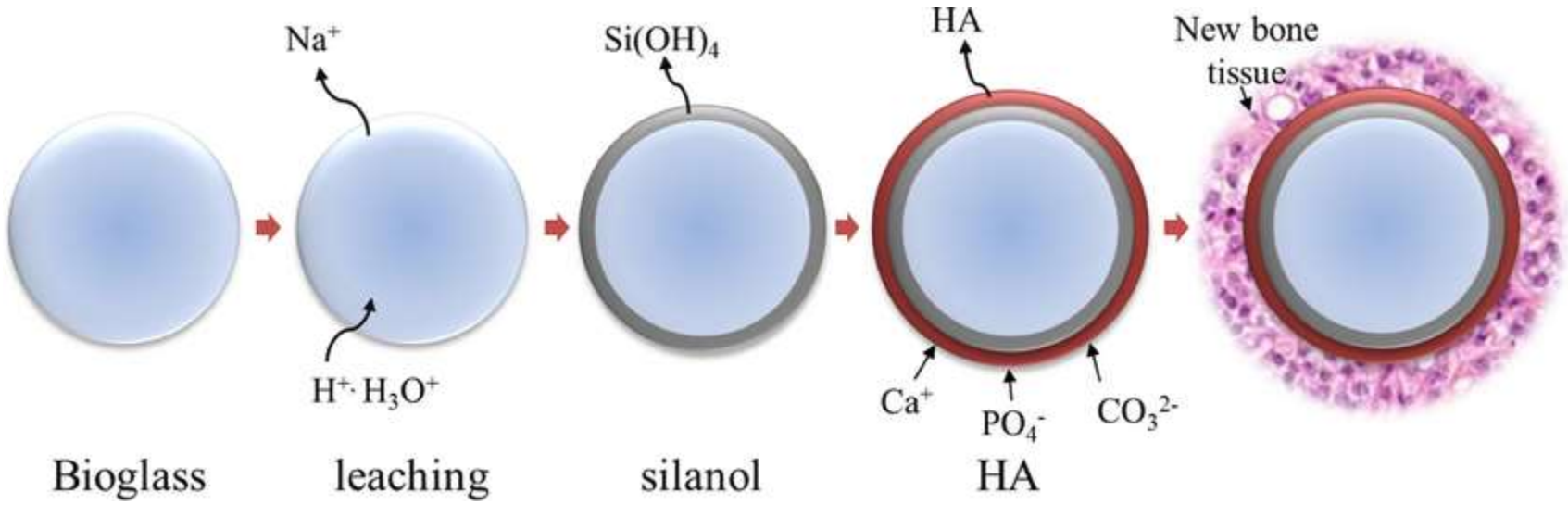
Stage I:

Alkali and alkaline earth ions are released into the fluid, and these ions in the glass structure are replaced by H^+ or H_3O^+ ions from the fluid. This reaction causes a rise in the local pH, which leads to the breaking of Si-O-Si bonds



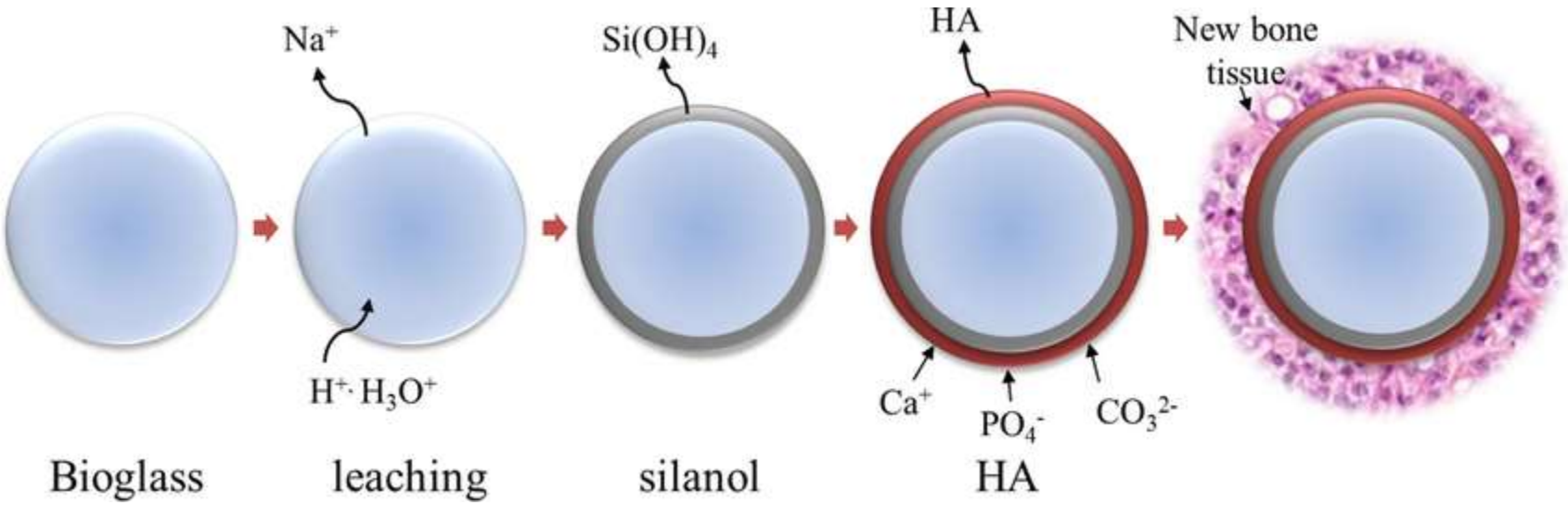
How does bioactive glass interacts with tissues?

Stage II: As a result, silicon is released into the fluid in the form of silanol ($Si(OH)_4$).



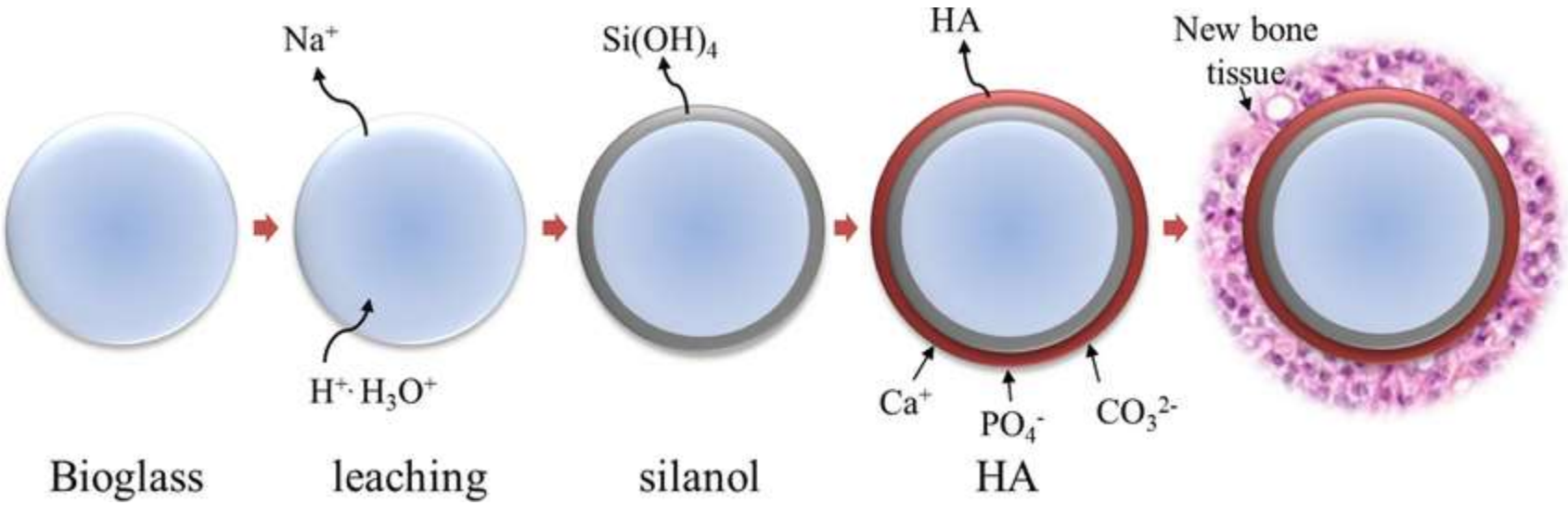
How does bioactive glass interacts with tissues?

Stage III: If the local pH remains below 9.5, $Si(OH)_4$ condenses and forms a polymerized silica gel layer on the glass surface. The open structure of the silica gel allows continued ion exchange between the glass and fluid.



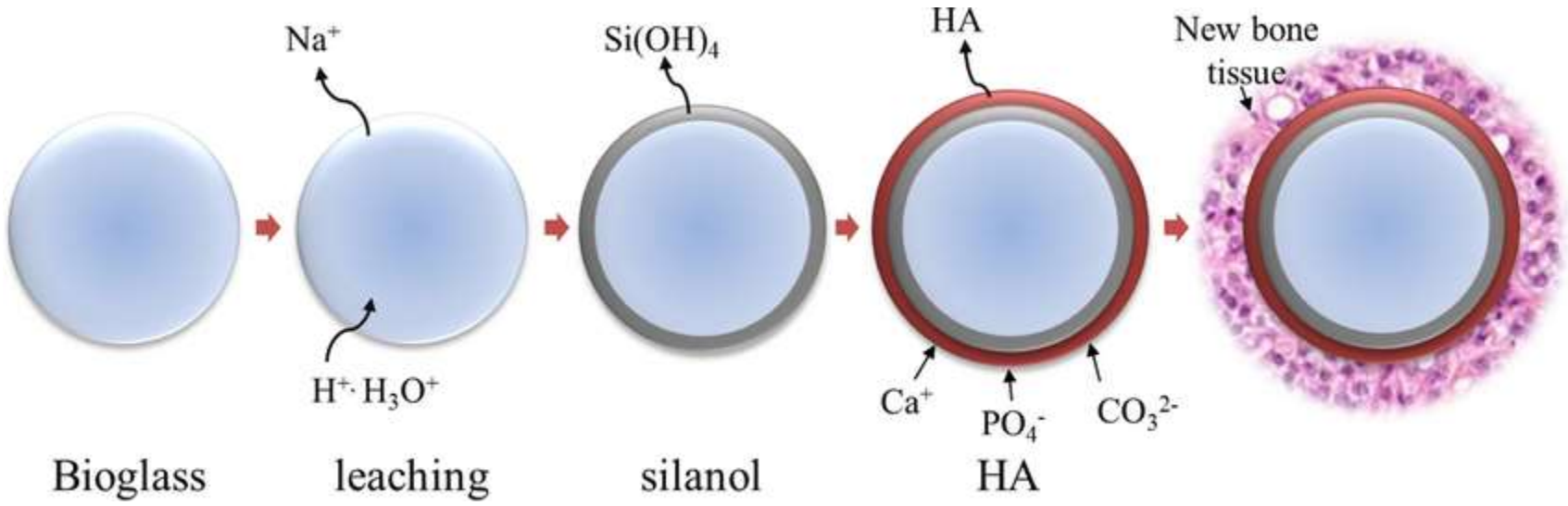
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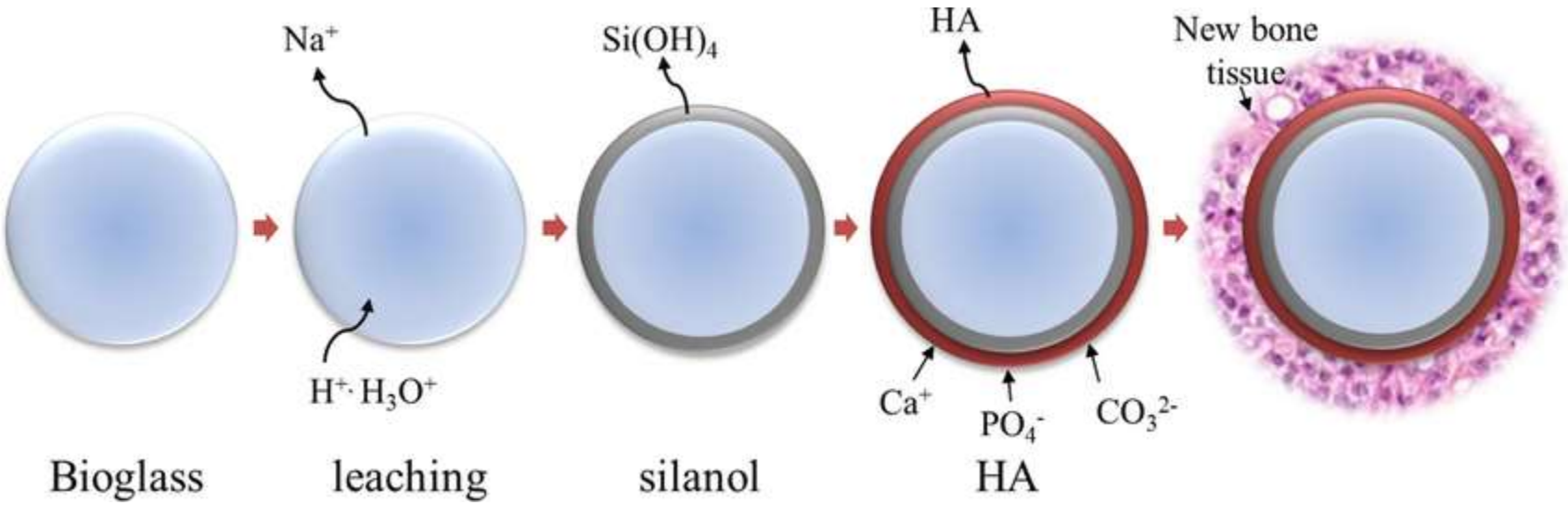
How does bioactive glass interacts with tissues?

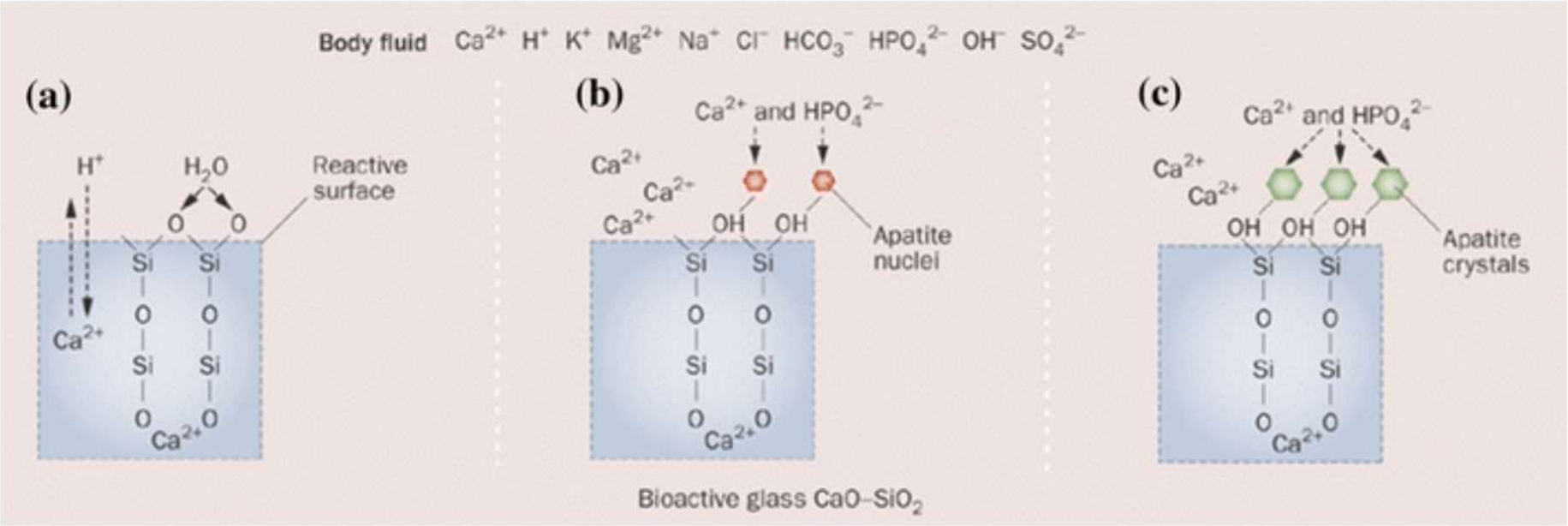
Stage IV: Calcium and phosphate ions diffuse out from the glass, and along with calcium and phosphate ions from the fluid, form an amorphous calcium phosphate layer on the silica gel.



How does bioactive glass interacts with tissues?

Stage V: As the silica gel and amorphous calcium phosphate layers thicken, carbonate species incorporated into the layer begin to crystallize, forming hydroxycarbonate apatite (HCA). HCA is chemically and structurally similar to the apatite mineral phase found in bone tissue.





The mechanism of bioactive glass surface modification upon exposure to body fluid (reproduced from Ref. [10] with permission)

Table 1. Composition and properties of bioactive glasses and glass–ceramics used clinically for medical and dental applications [1]

Composition (wt. %)	**45S5 Bioglass (NovaBone, Perioglass, NovaMin, Biogram)**	**S53P4 (AbminDent1, BonAlive)**
Na_2O	24.5	23
CaO	24.5	20
CaF_2	0	0
MgO	0	0
P_2O_5	6	4
SiO_2	45	53

Applications of bioactive glasses

Bone Grafts and Tissue Engineering: Bioactive glasses are often used as alternatives to bone grafts in orthopedic and dental surgery. They promote bone growth and help regenerate damaged or missing bone tissue. These glasses can be used in spinal fusions, joint replacements, and other bone repair procedures.

Dental Implants: Bioactive glasses can be incorporated into dental implants and coatings, enhancing osseointegration and supporting better integration with the surrounding jawbone. This increases the stability and long-term success of dental implants.

Wound Healing: Bioactive glass powders or fibers can be used in wound dressings to accelerate the healing process. They release ions like calcium and phosphate, which promote cell growth and tissue regeneration.

Jaw and Facial Reconstruction: Bioactive glasses are used in facial and jaw reconstruction surgeries, providing structural support while facilitating bone regeneration.

Drug Delivery: Bioactive glasses can be loaded with drugs or therapeutic agents and used as drug delivery systems. They release drugs slowly at the implant site, making them suitable for localized treatments and reducing the need for frequent drug administration.

Ear, Nose, and Throat (ENT) Surgery: Bioactive glasses have applications in ENT surgeries, particularly in middle ear and nasal cavity reconstructions.

Applications of bioactive glasses



Cavitary Defects

Diagnosis: The patient had a unicameral (single-chamber) bone cyst in the right ischium (part of the pelvis) located just below the acetabular component (socket) of a right total hip replacement. This cyst created a hollow or “cavitary” defect in the bone.

Treatment: At the time of revision arthroplasty, the cyst was debrided with the removal of all soft tissues to expose vital bone. The defect was grafted with a 50:50 mixture of **NovaBone** and locally harvested autograft.



Results: Three years after grafting, radiographs indicate filling of the site with dense bone tissue. No recurrence of the cyst, nor migration of the new acetabular cup has been noted (data on file). (pre-operative on left, 3-year post operative on right)

In short, this report describes a successful treatment for a bone cyst near a hip replacement, where bone grafting helped regenerate bone in the area with no complications after three years.

Applications of bioactive glasses



Maxillofacial Repair

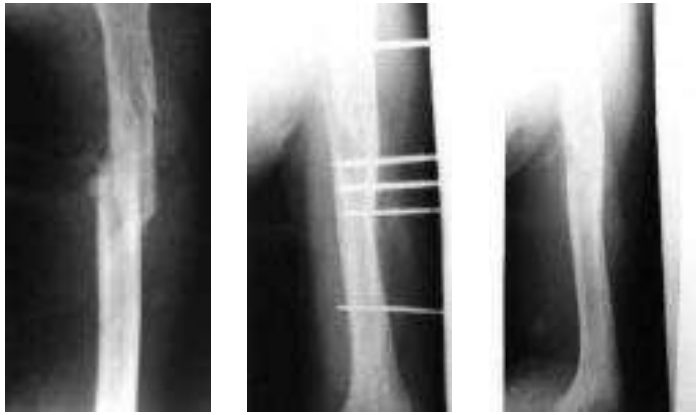
Diagnosis: Odontogenic keratocyst near left mandibular third molar, extending into the ramus

Treatment: Mandibular left first, second and third molars extracted. Cyst enucleated. Defect grafted with NovaBone-C/M and covered with a collagen membrane



Radiographic Results: The graft site, approximated by the dashed line, is filled with radio-dense tissue. No recurrence of cyst at six months apparent.”””””””””””””””

Applications of bioactive glasses



Fracture Repair

Diagnosis: Traumatic spiral fracture of the left humerus following an automobile accident.

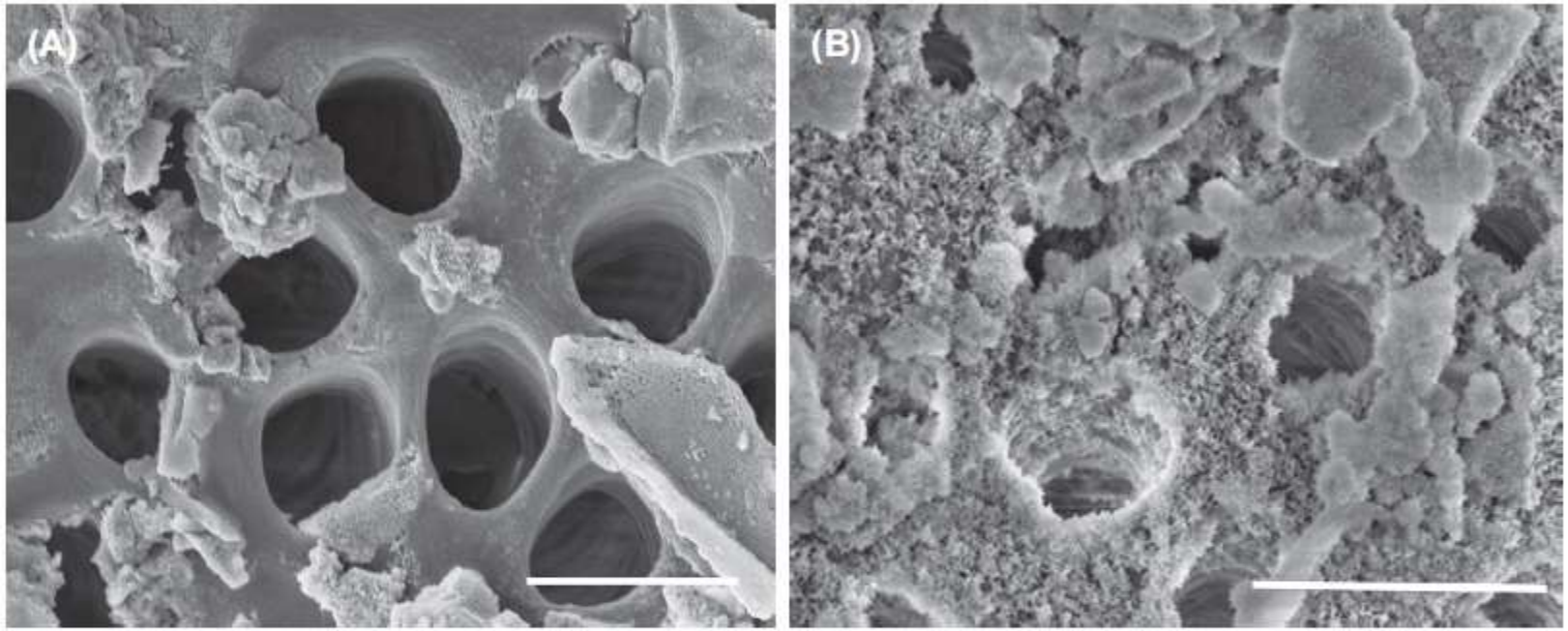
Treatment: Debridement and stabilization of the defect with a five-pin external fixator. The defect was grafted with 5cc of **NovaBone**.

Results: At four weeks, the graft area appears as a slightly radiodense area midshaft around the central pins. By six months, the fracture has healed and the fixator removed. (Data on file. (pre-operative on left; 4-weeks post-operative middle; 6-months post-operative on right)

Applications of bioactive glasses

Bioactive glass is not only used as a medical device or implant, it has also has an impact in consumer healthcare, with the largest commercial use of a bioactive material being in toothpaste.

Enamel and dentine of the tooth are very similar to bone, in that they contain HCA mineral and collagen. The pain felt by 35% of the adult population hypersensitivity is due to exposure of the microtubules of the dentine



• **Figure 1.3.4.10** Scanning electron microscopy images of human dentin treated with Sensodyne Repair and Protect (UK formulation) toothpaste that contains NovaMin (45S5) particles: (A) immediately after application of NovaMin in artificial saliva; (B) 24 h after application. Scale = 5 μm. (Modified from Earl, J.S., Leary, R.K., Muller, K.H., Langford, R.M., Greenspan, D.C., 2011. Physical and chemical characterization of dentin surface, following treatment with NovaMin® technology. J. Clin. Dent. 22, 62–67.)

NovaMin releases calcium and phosphate species during glass dissolution, which then form HCA on the dentine.

Manufacturing techniques

1. **Melt quench synthesis**

Conventional technique used by Hench

Melting of oxides at high temperatures

Dense scaffolds, potential risk of defects

2. **Sol-gel synthesis**

Conducted at low temperatures

Metal organic and metal salt precursors, forming gel by hydrolysis and condensation reactions

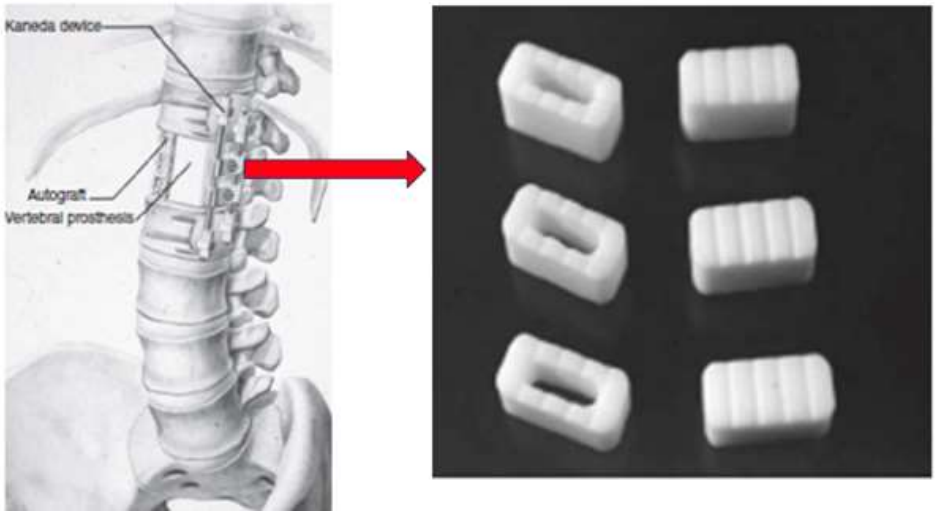
Higher control, higher porosity which leads to better degradation

https://youtu.be/Nk4vhi_LswM?t=180

Bioactive glass ceramics

Glass-ceramics are polycrystalline materials produced through controlled crystallization of base glass, producing a fine uniform dispersion of crystals throughout the bulk material.

Bioactive glass-ceramics

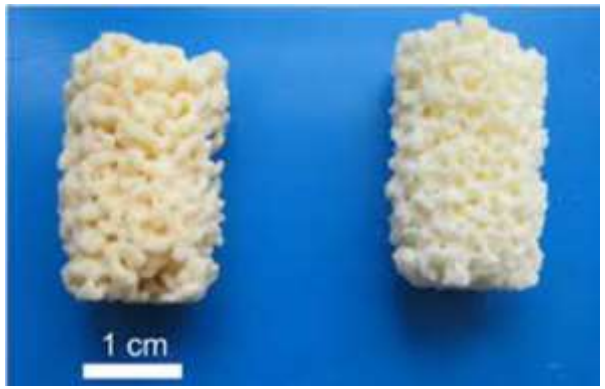


This novel material (as it were) contains the crystal phase of wollastonite and apatite ($CaOSiO_2$) within a glassy structure.

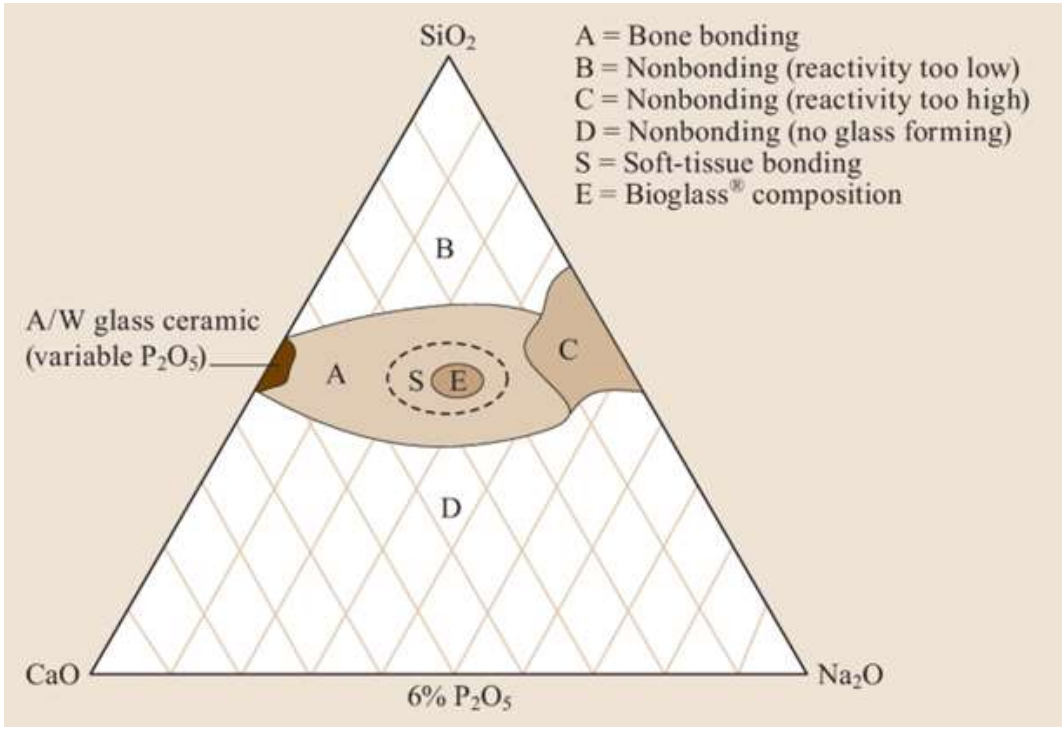
It has superior mechenical properties.

A/W glass ceramic: SiO_2–P_2O_5–CaO–MgO–CaF_2

P_2O_5 is added for nucleation.



The presence of both crystalline and amorphous regions is typical in glass-ceramics and gives A/W glass-ceramic a combination of beneficial properties, such as high fracture toughness, bioactivity, and partial resorbability.

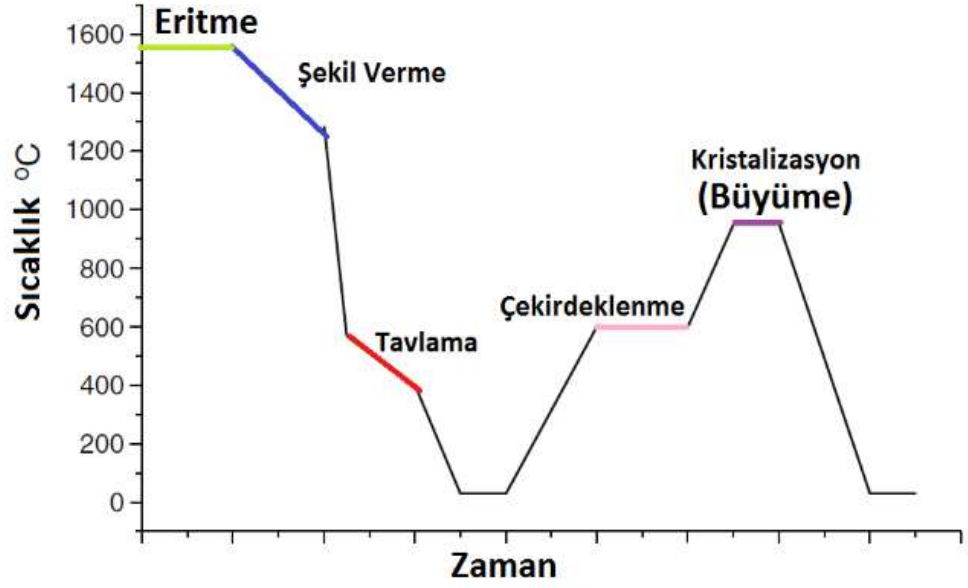


Bioactive glass-ceramics

Camlar kontrollü kristalizasyonla cam-seramiklere dönüştürülür.

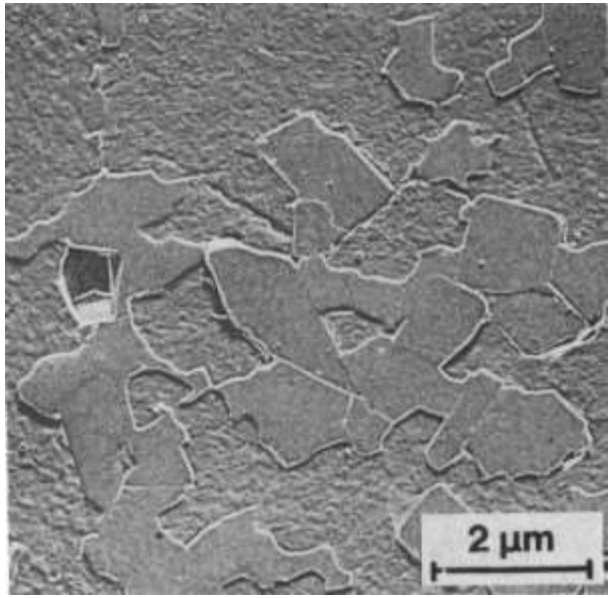
Kontrollü kristalizasyon şu adımlardan oluşur:

- **Çekirdeklenme**
- **Kristalizasyon**



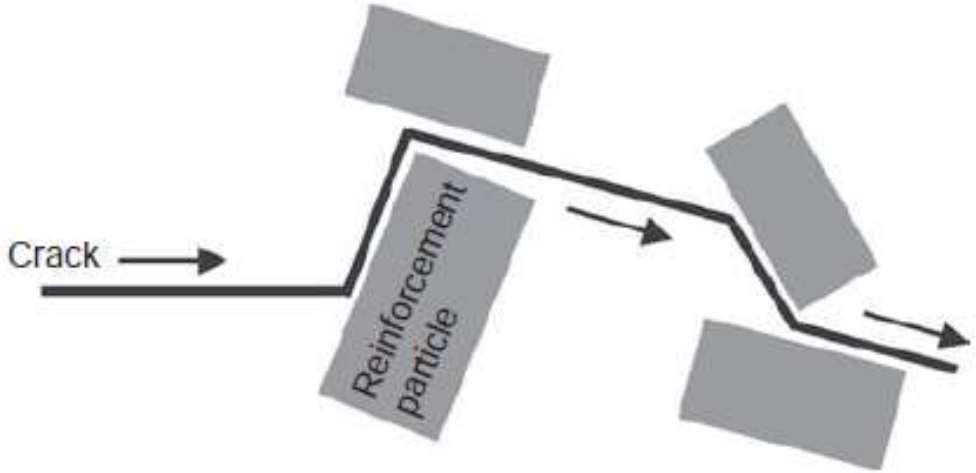
First, the glass is heated at a temperature range of 500–700°C to produce a large concentration of nuclei from which crystals can grow.

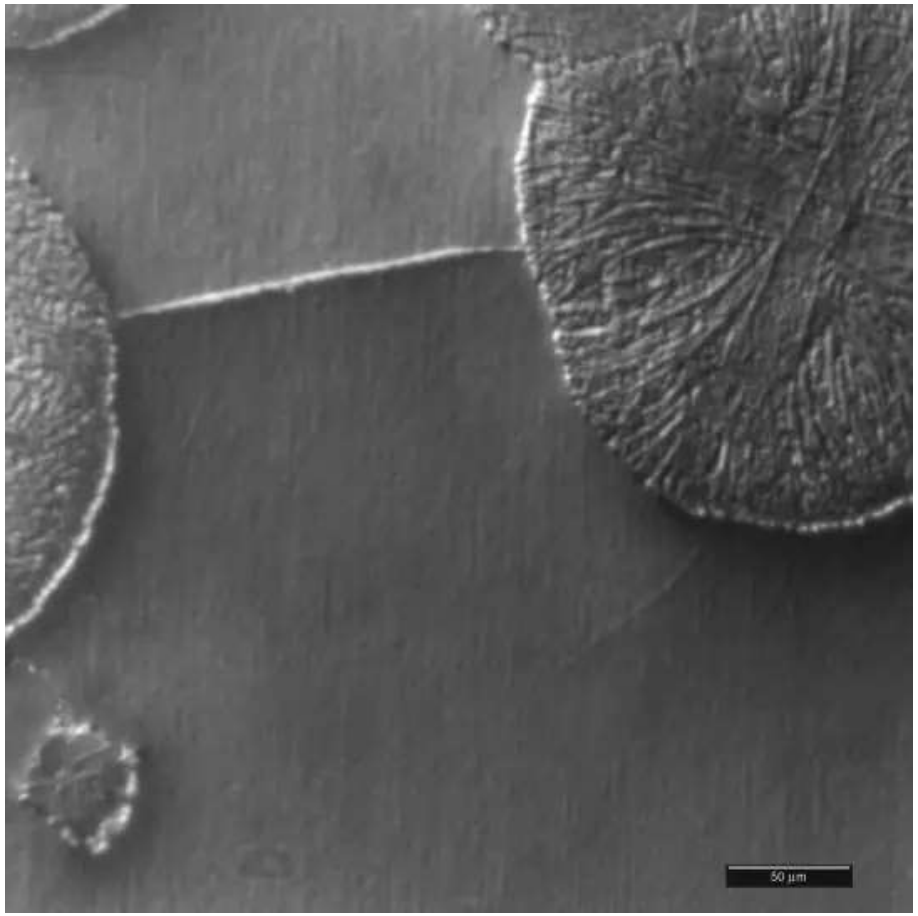
When enough nuclei are present to ensure that a fine-grained structure will be obtained, the temperature of the object is raised to a range of 600– 900°C to promote crystal growth.



Bioactive glass-ceramics

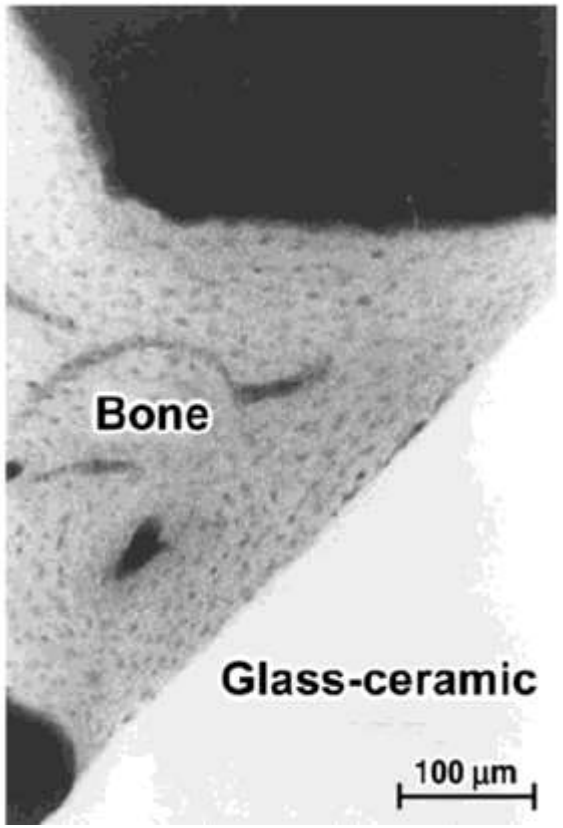
Ceramic crystallites induces crack deflection, which cosumes energy. This is reflected in increase in toughness.





Bioactive glass-ceramics

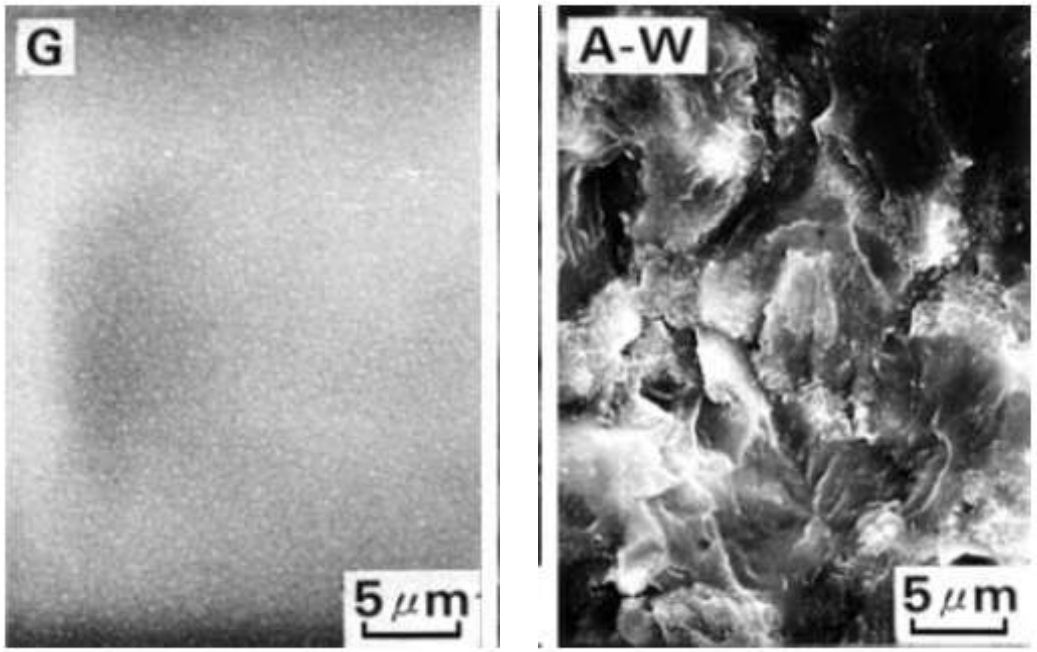
The response of living tissue to glass-ceramic A-W was examined by implanting rectangular samples 10 × 15 × 2 mm3 in size into the tibial metaphyses of a rabbit (Nakamura *et al*., 1985).

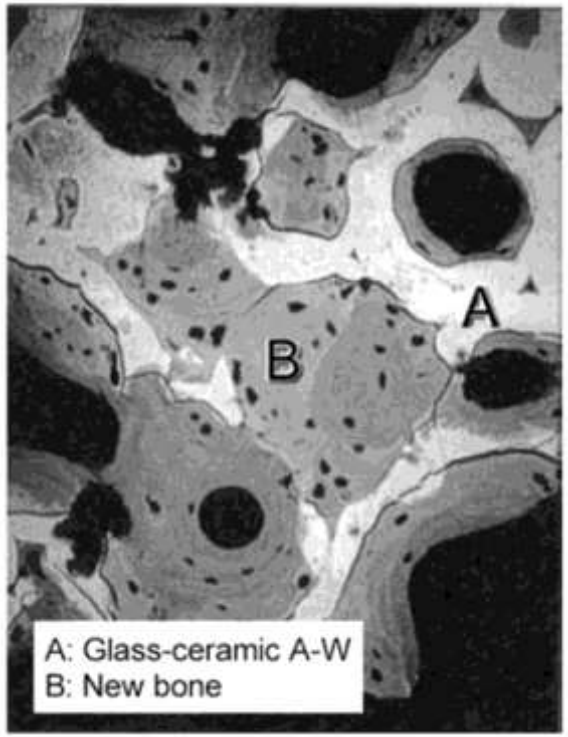


TEM image of the interface area
between rabbit tibia and glass-ceramic

Bioactive glass-ceramics

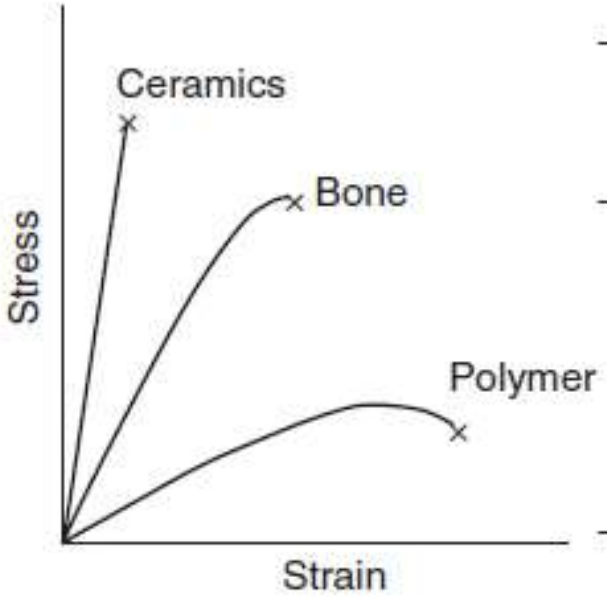
The high fracture toughness of A-W glass-ceramic is associated with its high fracture surface energy. While Glass G has a relatively smooth fracture surface, A-W glass-ceramic exhibits a much rougher fracture surface.





The porous form of A-W glass-ceramic partially dissolved within 12 weeks in a rabbit's tibia and fully integrated with newly grown bone.

Bioactive glass-ceramics



	Bending strength (MPa)	Compressive strength (MPa)	Young's modulus (GPa)
Bioglass®	42	–	35
Sintered hydroxyapatite	115–200	500–1000	80–110
Glass-ceramic A-W	215	1080	118
Cortical bone	50–150	100–230	7–30
Cancellous bone	10–20	2–12	0.05–0.5

–: not reported

Bioactive glass-ceramics

Özellik	Biyocam 45S5	HA Seramik	Cam-Seramik Ceravital	Cam-Seramik A/W
Yoğunluk ($g.cm^{-3}$)	2.66	3.16	–	3.07
Vickers Sertliği (Hv)	460	600	–	680
Elastisite (Young) Modülü (GPa)	35	80-110	100-150	218

Calcium Phosphate Ceramics

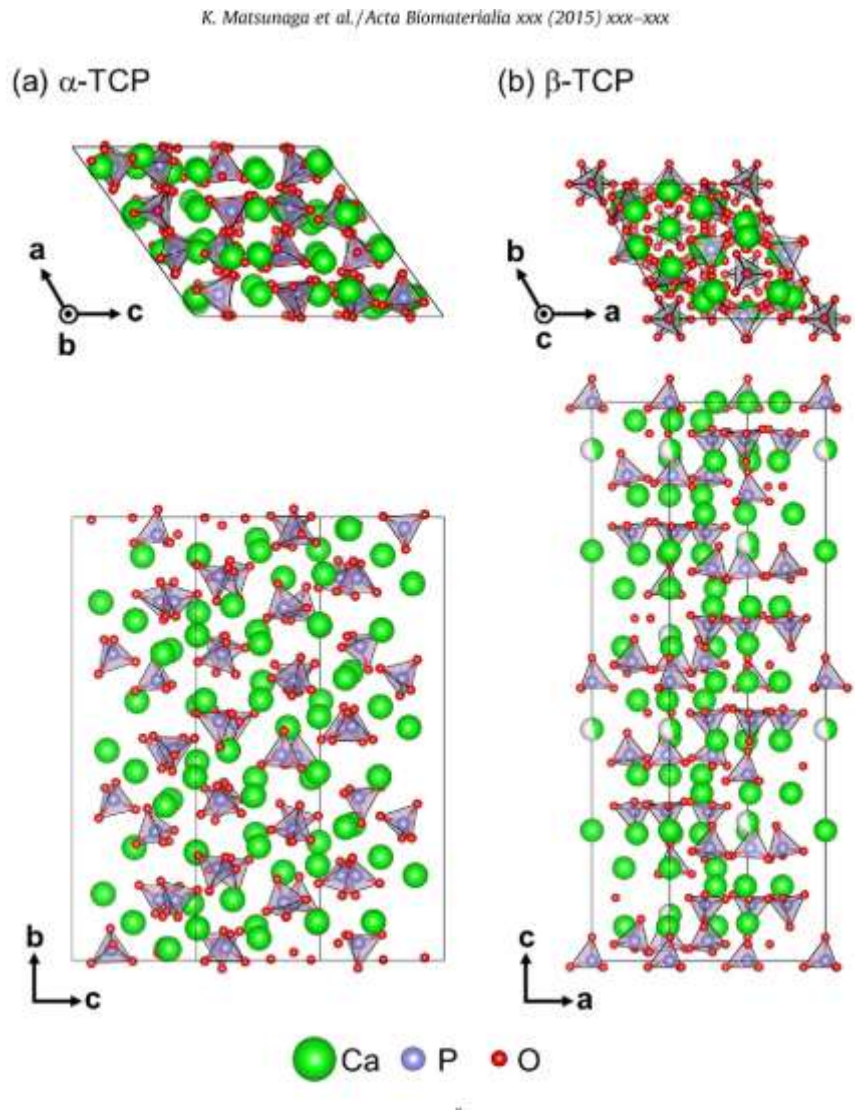
Kemiğe benzer implant malzemelerdir. Bileşimlerinin kemiğin mineral fazına benzer olması nedeniyle kemikle uyumludur.

Bileşiğin Adı	Kısaltma	Formül	Ca:P oranı
Tetrakalsiyum fosfat	TetCp	$Ca_4O(PO_4)_2$	2.0
Hidroksiapatit	OHAp	$Ca_{10}(PO_4)_6(OH)_2$	1.67
Amorf kalsiyum fosfat	ACP	$Ca_{10-X}H_{2X}(PO_4)_6(OH)_2$	
Trikalsiyum fosfat (α, β, γ)	TCP	$Ca_3(PO_4)_2$	1.50
Oktakalsiyum fosfat	OCP	$Ca_8H_2(PO_4)_6.5H_2O$	1.33
Dikalsiyum fosfat dihidrat	DCPD	$CaHPO_4.2H_2O$	1.0
Dikalsiyum fosfat	DCP	$CaHPO_4$	1.0
Kalsiyum fosfat (α, β, γ)	CPP	$Ca_2P_2O_7$	1.0
Kalsiyum fosfat dihidrat	CPPD	$Ca_2P_2O_7.2H_2O$	1.0
Heptakalsiyum fosfat	HCP	$Ca_7(P_5O_{16})_2$	0.7
Tetrakalsiyum fosfat diasit	TDHP	$Ca_4H_2P_6O_{20}$	0.67
Kalsiyum fosfat monohidrat	MCPM	$Ca(H_2PO_4)_2.H_2O$	0.5
Kalsiyum metafosfat (α, β, γ)	CMP	$Ca(PO_3)_2$	0.5

Bone graft substitutes produced from porous calcium phosphates usually consist of either HA or β-TCP, or a mixture of these two phases to produce biphasic calcium phosphates (BCPs). The relative solubility of β-TCP is much higher than that of HA. Both of these materials are osteoconductive, providing a surface that encourages new bone formation, with their major difference being the rate of degradation in vivo

Tetrakalsiyum fosfat, $Ca_4P_2O_9$,
Amorf kalsiyum fosfat
Trikalsiyum fosfat, (TCP), $Ca_3(PO_4)_2$,
Hidroksiapatit, (HA), $Ca_5(PO_4)_3OH$

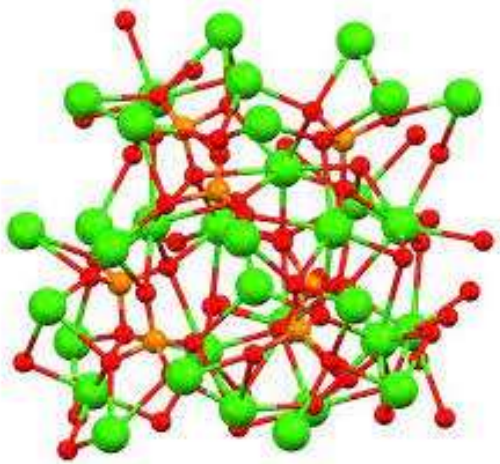
Çözünebilirlik

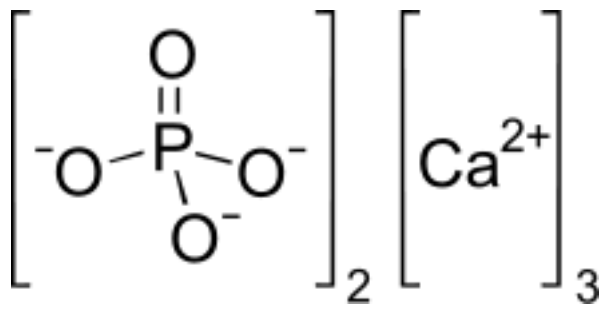


In cases where rapid resorption of the bone graft is required, the use of β-TCP could be preferred to HA, but the challenge could be that the β-TCP resorbs faster than the rate of bone formation.

Hydroxyapatite is bioactive but **not bioresorbable**

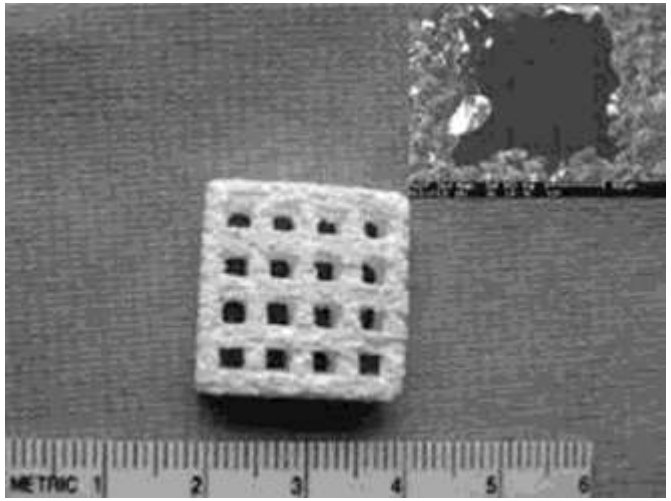
Beta-tricalcium phosphate (β-CaP) has a slower dissolution rate than alpha-tricalcium phosphate (α-CaP). This difference in dissolution rate is due to their distinct crystal structures, which influence how they interact with bodily fluids.





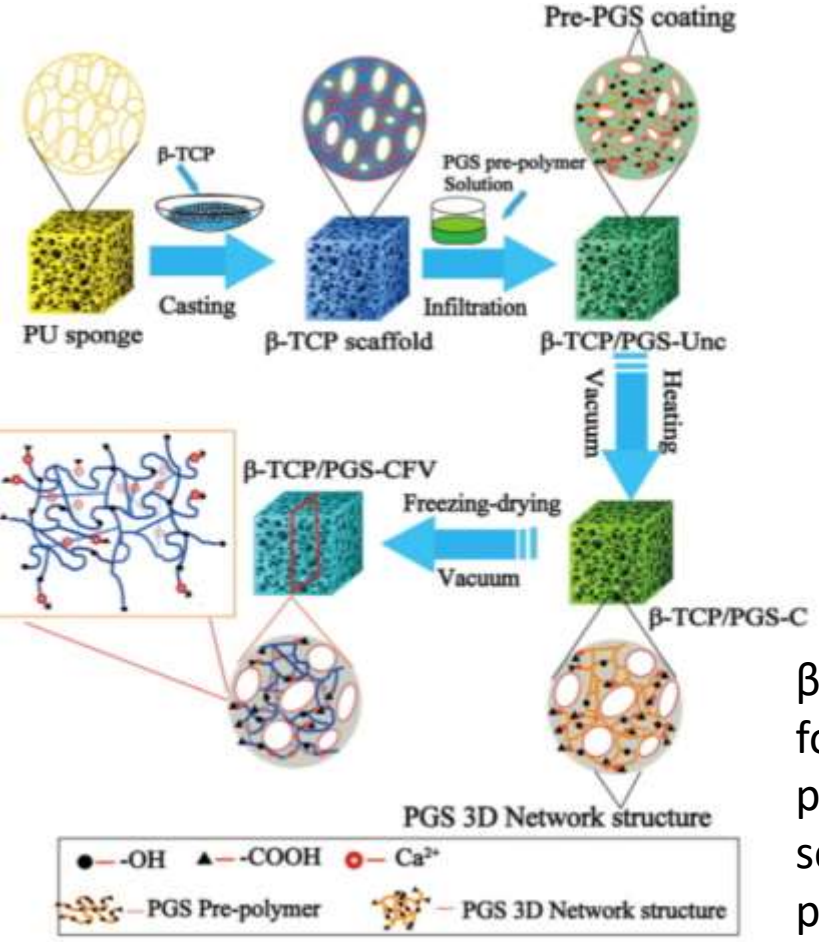


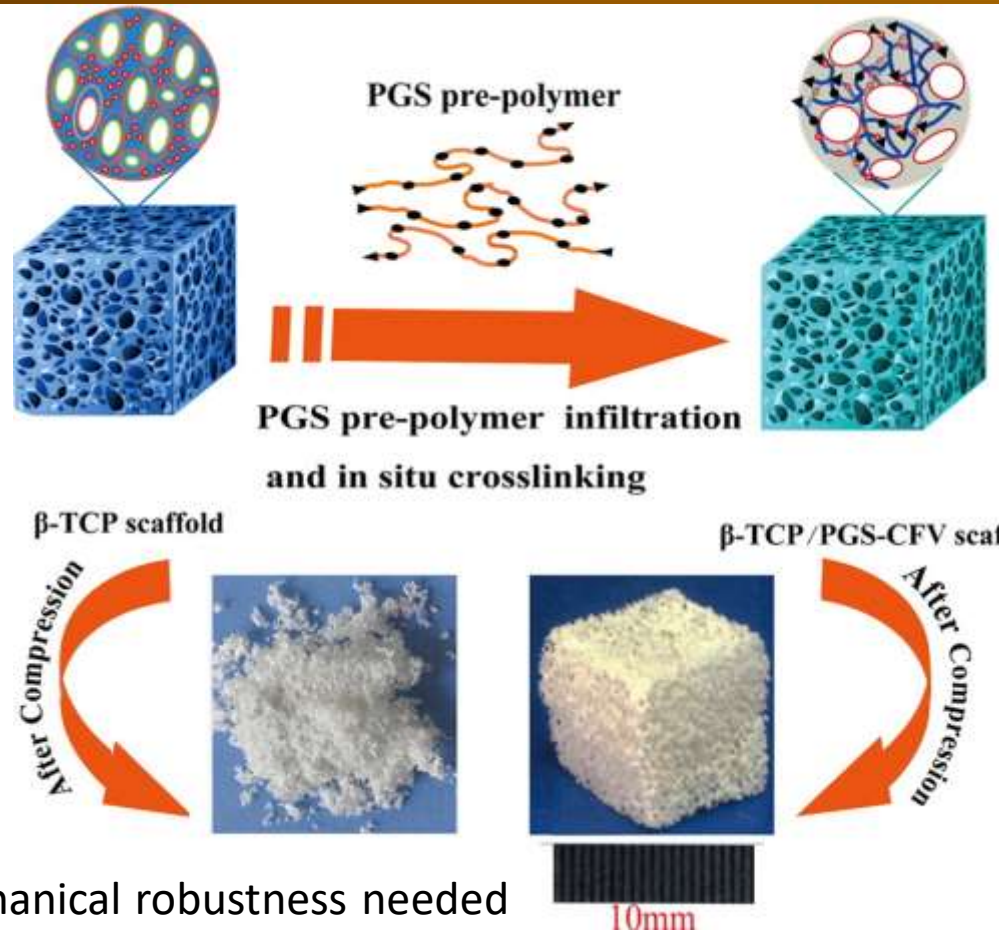




Porous Calcium Phosphate Ceramics

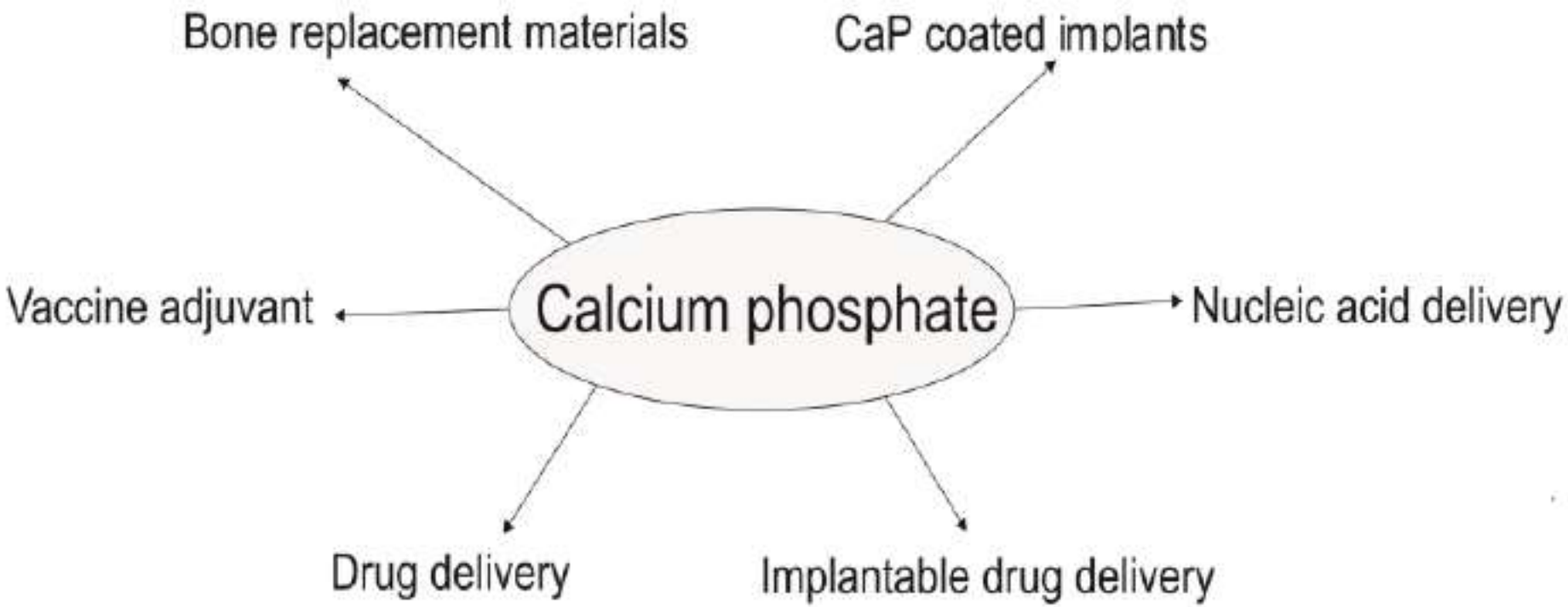
1. β-TCP Scaffold Formation Using a Polyurethane Sponge Template
2. PGS Impregnation and Crosslinking
3. Freeze-Drying and Vacuum Processing



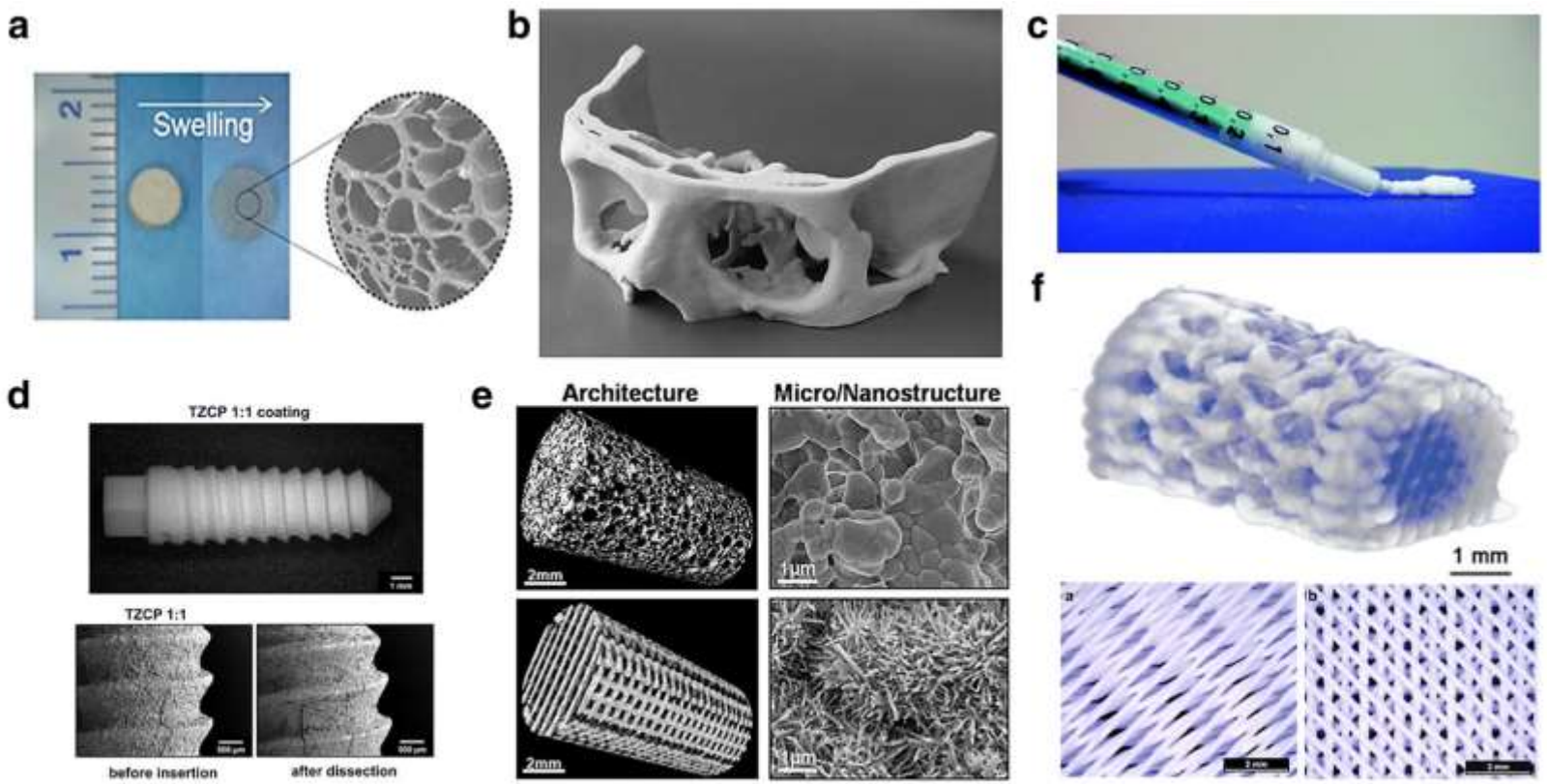


β-TCP lacks the flexibility and mechanical robustness needed for load-bearing applications, while PGS alone does not provide the osteoconductivity of β-TCP. Thus, this composite scaffold leverages the best of both materials, making it a promising material for advanced bone grafts and implants.

Calcium Phosphate Ceramics



Calcium Phosphate Ceramics

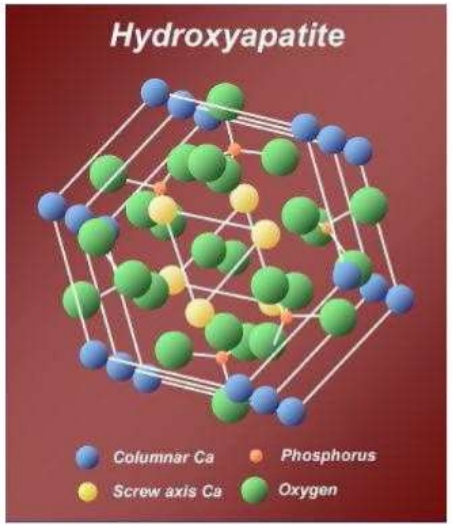


Calcium phosphate based applications. (**a**) WH incorporated hydrogel scaffold [116, 176]. (**b**) Cranial segment made of tetracalcium phosphate and β-TCP [177]. (**c**) The injectable paste included calcium phosphate nanoparticles [178]. (**d**) Mixed zirconia calcium phosphate deposited on dental implant [179]. (**e**) 3D printed calcium-deficient HAP scaffolds [180]. (**f**) 3D printed calcium phosphate cement [181]

Hidroksiapatit (HA)

Kemikler ve diş minesi; HA, protein, organik maddeler (kolajen) ve su içeren kompozitlerdir.

Kemik kolajen lifleri içerisine gömülmüş, 25-50 nm uzunluğunda inorganik (HA) içeren, doğal bir kompozittir.



Ca/P oranı	1,67
Yoğunluk	3,16 g/cm^3
Elastisite Modülü	40-117 GPa
Kafes	Hekzagonal Rombik

Apatit, $M_{10}(RO_4)X_2$ genel formülüne sahip bileşiklere verilen isimdir. R fosfor, M genellikle Ca olmak üzere çeşitli metaller, X ise hidroksit veya halojenlerdir (F).

HA kemiğin inorganik bileşenine benzer olup, $Ca_{10}(OH)_2(PO_4)_6$ formülündedir.

Apatit $M_{10}(RO_4)X_2$ genel formulünde M, R ve X yerine gelebilecek elementler	
Grup	Element
M	Ca, Sr, Ba, Cd, Pb, Mg, Na, K
R	P, CO_2, As, Si, Ge, Cr, B, V
X	OH, CO_2, O, F, Cl, Br

Summary

Bioinert Ceramics

- **Example: Alumina (Al_2O_3)**
- **Properties: Bioinert ceramics maintain their physical and mechanical properties within the body.**
- **Characteristics: They are passive materials and do not chemically interact with surrounding tissues, making them suitable for applications where stability is key.**

Summary

Bioactive Ceramics

- **Examples: Bioglass, Glass Ceramics, Hydroxyapatite (HA)**
- **Properties: Bioactive ceramics form strong chemical bonds with tissues, promoting secure attachment to bone.**
- **Characteristics: They are often porous, allowing body fluids and air to pass through, which supports cell growth and tissue integration.**

Summary

Bioresorbable/Biodegradable Ceramics

- **Examples: Calcium Phosphates**
- **Properties: These ceramics gradually dissolve and are absorbed by the body over time.**
- **Characteristics: Used in applications where temporary support is needed, such as in bone healing, after which the material is replaced by natural tissue.**

Bibliography

https://doi.org/10.1016/j.jnoncrysol.2015.03.022

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